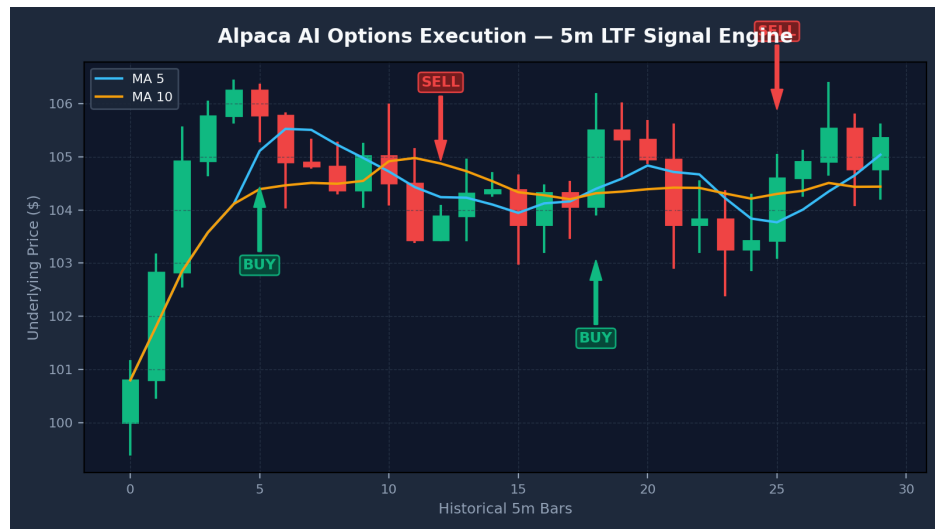


Alpaca AI Options RL Execution Pipeline

Instrument-Agnostic Contextual Q-Learning & Multi-Horizon Meta-Learner



Discipline-Driven Execution: Eliminates over-trading greed with DeepScalper reward shaping.

Two-Tier Framework: Tier 1 HTF Meta-Learner + Tier 2 Dual-Branch Q-Executor.

0% Lookahead SNR Engine: Vectorized Causal K-Means volume profile zone snapshots.

1:1 Framework Parity: PyTorch and Keras multi-horizon execution server parity.

Cross-Symbol Testing: USD Equity & ETF universe (GLD, SPY, QQQ, TLT, etc.).

1. Two-Tier Architecture & Workflow Pipeline

Decoupled multi-horizon context scoring and zero-interference execution.

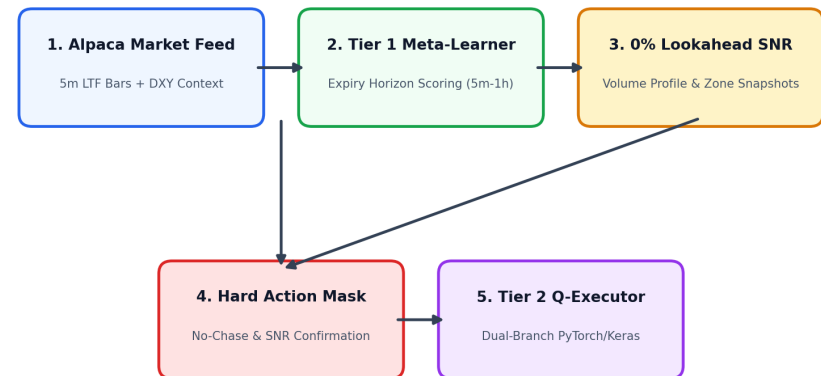
Tier 1 — Signal Meta-Learner:

- Evaluates 24-bar price statistics (MFE, MAE, reversal timing) for conviction score [0.0 - 1.0].
- Predicts optimal option expiry duration (5m, 15m, 30m, 1h) from MTF RSI & DXY context.
- 6 decoupled auxiliary regression heads with private LayerNorm projections.

Tier 2 — Dual-Branch Ensemble Q-Learner:

- Microstructure + MTF Alignment gated fusion layer.
- 5-action space: WAIT, CALL, PUT, TP_HALF, CLOSE.
- Full 1:1 functional parity between PyTorch and Keras engines.

AXE Genesis Two-Tier Reinforcement Learning Pipeline

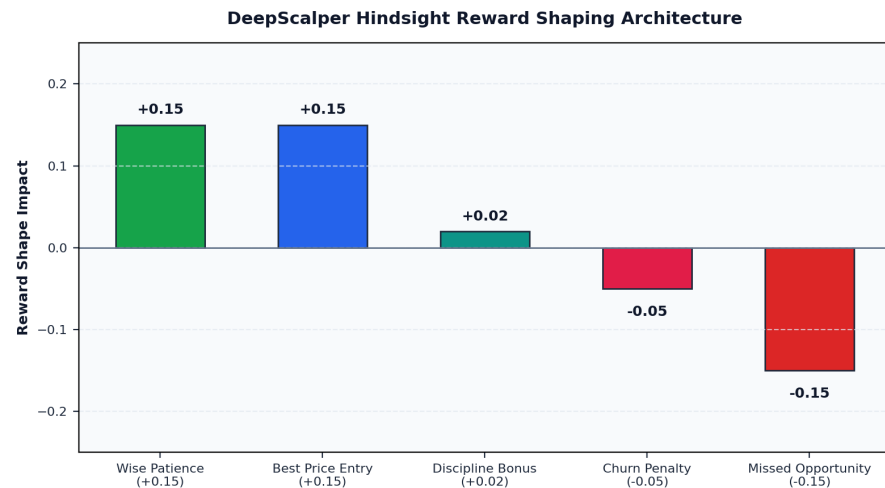


2. DeepScalper Hindsight Reward Shaping & Wise Patience

Aligning execution discipline against opportunity cost without distortion.

Key Behavioral Reward Mechanisms:

- **Wise Patience (+0.15)**: Positively rewards WAIT when standing down avoids adverse loss.
- **Missed Opportunity (-0.15)**: Penalizes WAIT if an unmasked, high-conviction setup played out.
- **Best Price Entry (+0.15)**: Rewards entries with minimal drawdown exposure ($\leq 0.5\%$).
- **Discipline Bonus (+0.02)**: Constant reward for WAIT during low-conviction regimes.
- **No-Chase Masking**: Enforces SNR proximity & volume delta before unmasking CALL/PUT.



3. Data Integrity & Real Market Engine

Strict zero-lookahead S&R; detection and dynamic instrument scale normalization.

Zero-Lookahead SNR Zone Detection:

- Vectorized support_resistance.py module (detect_snr_levels_sequential + create_clustered_zones_sequential).
- Zone snapshots generated dynamically per bar strictly using causal historical slices: `df.iloc[:idx+1]`.
- Completely eliminates lookahead bias and static quantile placeholders across all 4 evaluation phases.

Real Volume Profiles & Hard Action Masking:

- `_make_exec_ctx` computes real candle volume profiles (buy volume vs sell volume) directly from market feeds.
- `HardActionMask` enforces zone proximity and volume delta confirmation before unmasking CALL/PUT entry actions.

Dynamic Instrument Metadata Normalization:

- `get_instrument_metadata` automatically maps ETF/Equity underlyings (GLD, SPY, QQQ, TLT, SLV, AAPL, etc.) to `pip_size=0.01`.
- Resolves 100x scale distortions between Equities ($\$0.01 = 1 \text{ cent} = 1 \text{ pip}$) and FX/Crypto.

4. Cross-Symbol Out-of-Sample Evaluation

Multi-symbol walk-forward backtesting across standardized option underlyings.

Cross-Symbol Benchmark Loop:

- Evaluates 40,000 historical 5m bars (~2 years of market data) per symbol across ETF/Equity universe.
- Out-of-sample holdout test set (20% split, ~8,000 bars) walk-forward execution.

Scientific Notation Micro-Loss Tracking:

- All training logs standardized to scientific notation ($\cdot 4e$) to track micro-losses across auxiliary heads.
- Confirms loss convergence for Q-values, directional strength, pips proxy, and reversal risk heads.

Expiry Horizon Selection Summary:

- Classifies optimal duration across 5m (1 bar), 15m (3 bars), 30m (6 bars), and 1h (12 bars) option expiries.
- Reports directional win rates, maximum win/loss streak bounds, and meta-learner expiry alignment rates.

5. Summary & Key System Strengths

Robust, production-grade RL execution architecture built for Alpaca AI.

Key Achievement	Implementation Details
Two-Tier RL Architecture	Tier 1 Meta-Learner (Context/Expiry) + Tier 2 Dual-Branch Q-Executor (Entry/Exit).
PyTorch / Keras Parity	100% architectural & reward parity across both major deep learning frameworks.
DeepScalper Reward Shaping	Wise patience bonuses and hindsight opportunity penalties to train disciplined policy.
0% Lookahead SNR Engine	Sequential K-Means volume profile zone snapshots operating on causal price slices.
Alpaca Market Pipeline	Real-time MTF bar fetching with hard timeout guards and dynamic DXY inversion features.

Submitted for lablab.ai Alpaca AI Hackathon — Built with PyTorch, Keras, Alpaca API & Matplotlib.